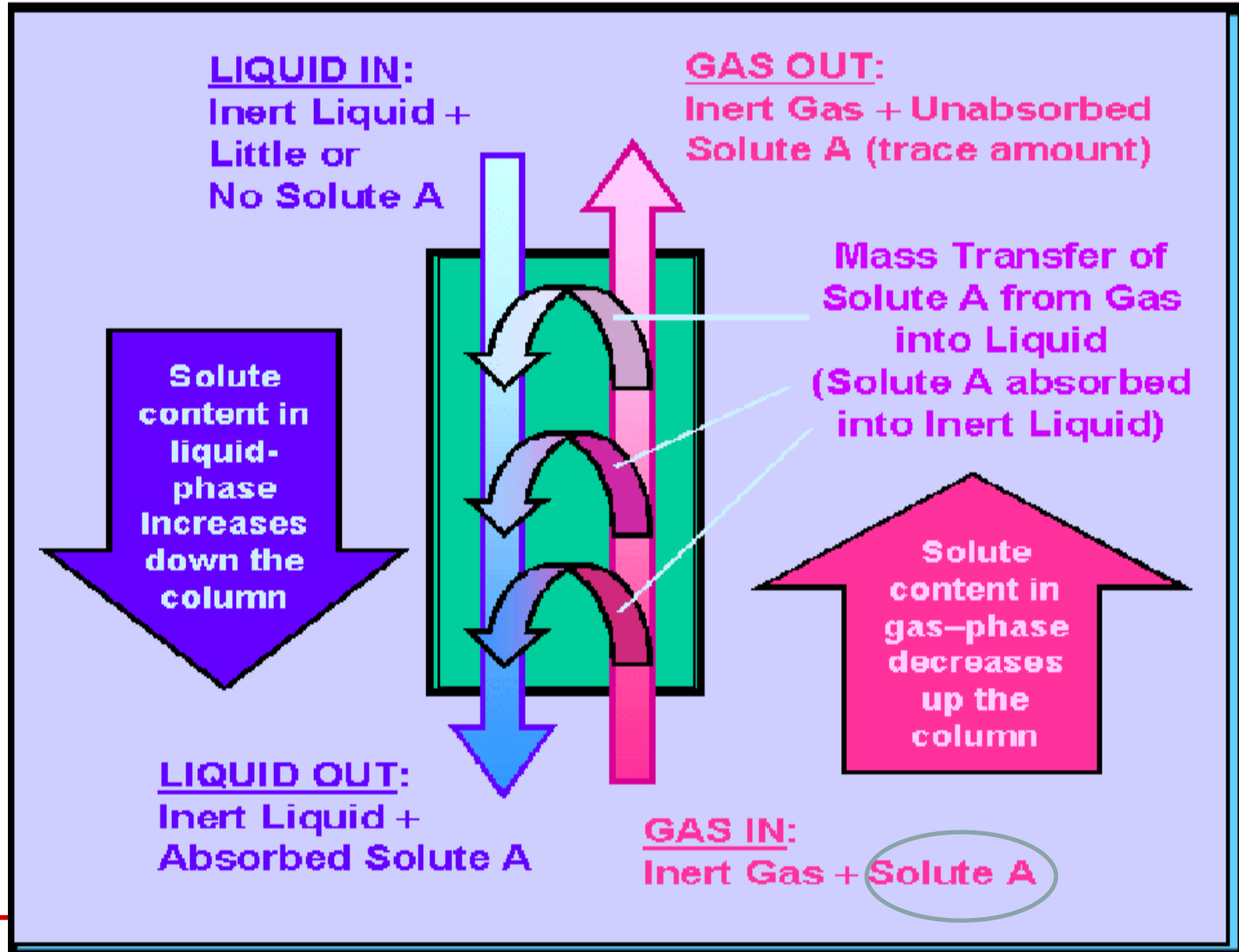
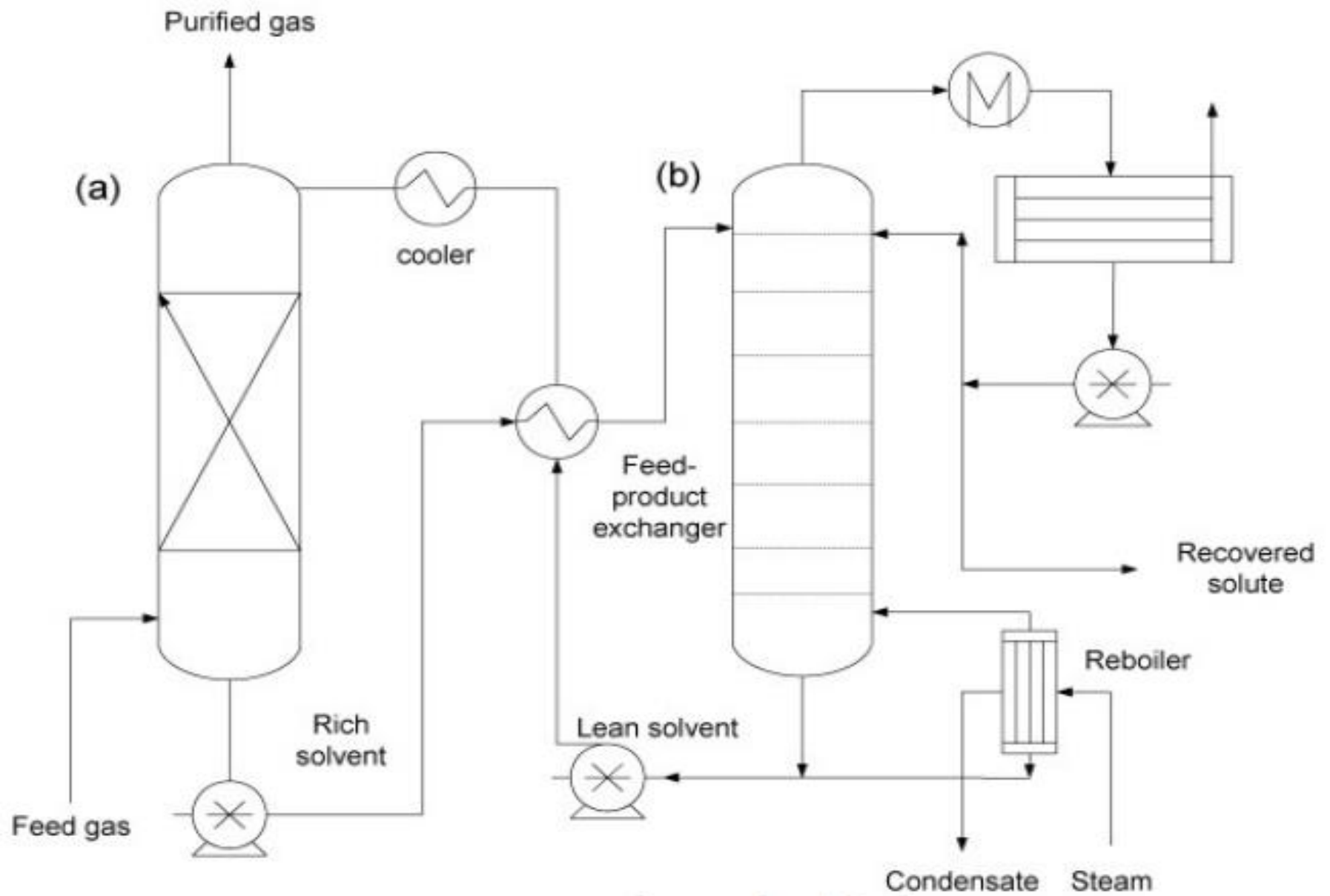


Chapter Five: Gas Absorption



- ❖ Gas absorption is a process in which one or more soluble components (**solutes**) are removed from a gas phase by contact with a liquid phase (**solvent**) in to which the components of interest dissolve.
- ❖ In general the absorbed gas is then removed from the solvent, and the solvent liquid stream is subsequently returned to the system.
- ❖ The solvent recovery process is called **stripping**. Stripping is also employed when volatile components have to be removed from a liquid mixture. The stripping agent is either a gas or super heated vapor.
- ❖ In most common units the solvent enters the top of the absorber/stripper and flows down wards **counter current** to the rising gas stream.
- ❖ The two phases mix and contact one another, either on **plates or Packings**.



❖ Solute transferred from

- *gas phase to the solvent--- Gas Absorption.*
- *Liquid to gas --- stripping*

➤ The inert gas in the gas mixture is called “**carrier gas**”.

➤ In the absorption process of ammonia from **air-ammonia** mixture by water, air is carrier gas, ammonia is “**solute**” and water is **absorbent**.



Absorption operation is of two types; Physical and Chemical.

Physical process:

- Acetone can be recovered from an acetone–air mixture by passing the gas stream into water in which the acetone dissolves while the air passes out.
- Ammonia may be removed from an ammonia–air mixture by absorption in water.
- Other examples

CO ₂ and water	Acetylene and acetic acid
CO and water	NH ₃ and acetone
H ₂ S and water	Ethane and carbon disulfide
NH ₃ and water	N ₂ and methyl acetate
NO ₂ and water	NO and ethanol
- Physical absorption relies on the solubility of a particular gas in a liquid.
- The solubility is often quite low; consequently, a relatively large amount of liquid solvent is needed to obtain the required separation

Chemical:

Chemical absorption relies on reaction of a particular gas with a reagent in a liquid.

- Oxides of nitrogen are absorbed in water to give nitric acid
- Carbon dioxide is absorbed in a solution of sodium hydroxide, a chemical reaction occurs, the nature of which influences the actual rate of absorption
- Other examples
 - CO_2 / H_2S and aqueous ethanol amines
 - CO_2 / H_2S and aqueous hydroxides
 - CO and aqueous Cu ammonium salt
 - SO_2 and aqueous dimethyl aniline
 - HCN and aqueous NaOH
 - HCl / HF and aqueous NaOH
- Such absorption can often be quite high;
 - consequently, a smaller amount of liquid solvent/reagent is needed to obtain the required separation. However, the reagent may be relatively expensive, and it is often desirable to regenerate when possible.

General design considerations

1. Entering gas composition, flow rate, temperature and pressure
 2. Desired degree of separation
 3. Choice of solvent:
 - High gas solubility
 - Low volatility
 - Low corrosiveness
 - Low cost
 - Low viscosity
 - Non-toxic, non-explosive
 4. Minimum solvent flow rate -> actual solvent flow rate
 5. Number of equilibrium stages
 6. Heat/cooling requirements
 7. Type of absorber
 8. Geometrical parameters of the system/economy
- Design objective**
- Thermophysical characteristics**
- Basic design elements**
- Technical specifications**

Choice of solvent for Absorption

The *following properties* are the important consideration, in choice of solvent for absorption.

1. *Gas solubility*

- The gas **solubility** should be high, thus *increasing the rate of absorption* and *decreasing the quantity of solvent required*.

2. *Volatility*

- The solvent should have a **low vapor pressure** since the gas leaving an *absorption operation* is ordinarily saturated with the *solvent* & much may be thereby be *lost*. If necessary, a *second, less volatile liquid* can be used to recover the **evaporated portion** of the first.

3. Corrosiveness

- ❖ The materials of construction required for the equipment should not be *unusual or expensive*.

4. Cost

- ❖ The solvent should be *inexpensive*, so that losses are *not costly and should be readily available*

5. Viscosity

- ❖ *Low viscosity* is preferred for reasons of rapid *absorption rates*, *improved flooding characteristics* in absorption towers, *low P drops* on *pumping* and *good heat transfer* characteristics

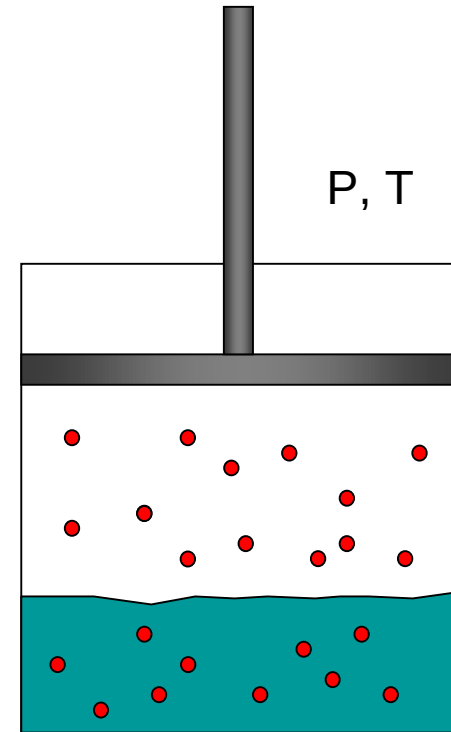
- The solvent if possible should be
 - *non toxic*
 - *non-flammable*
 - *chemically stable &*
 - *should have a low freezing point.*

Temperature:

- For the determination of **driving force** in any mass transfer operation, the solubility of a species in a solvent, i.e., **equilibrium distribution between phases** is important.
- With the increase in temperature, solubility of a gas in liquid decreases.
- Hence, absorption is done at lower temperature.
- On the contrary, **desorption is done at higher temperature.**

Thermodynamics of the absorption process

- Imagine a system where we have a binary mixture of two components **A** and **B**, one (A) is volatile and the second one (B) is a non-volatile solvent.
- The mixture coexists with an **equilibrium gas of almost pure volatile component**, which has pressure P .
- So if we place a piston on the top of the system, exerting pressure P , the system will be in equilibrium.
- The concentration of the volatile component A in solvent B is called **solubility of A in B** at P, T . This characteristics can be measured for a range of binary systems under different P, T conditions.
- In general **solubility decreases with temperature**. It is natural to guess that if we heat the system up, the energy supplied in the system will be used to evaporate more of the volatile component A.



Differences b/n Distillation and Absorption

Distillation

- ❖ The vapor has to be produce in each stage by partial vaporization of the liquid at its **boiling point**
- ❖ Diffusion of molecules exists in both direction

Absorption

- The liquid is **well below its boiling point**
- Gas molecules are diffusing in to the liquid, the movement in to the **reverse is negligible.**
- The ratio of liquid to gas flow rate is greater in absorption than distillation

Equilibrium Between Liquid and Gas

- When two phases are brought into contact they **eventually reach equilibrium**, water in contact with air evaporates until the air is **saturated with water vapour**, and the air is absorbed by the water until it becomes saturated with the individual gases.
- In any mixture of gases, the **degree to which each gas is absorbed** is determined by its **partial pressure**.

At a given temperature and concentration, each dissolved gas exerts a **definite partial pressure**.

Three types of gases may be considered from this aspect

- **very soluble one**, such as ammonia
- **moderately soluble one**, such as sulphur dioxide
- **slightly soluble one**, such as oxygen

Table 12.1. Partial pressures and concentrations of aqueous solutions of gases at 303 K

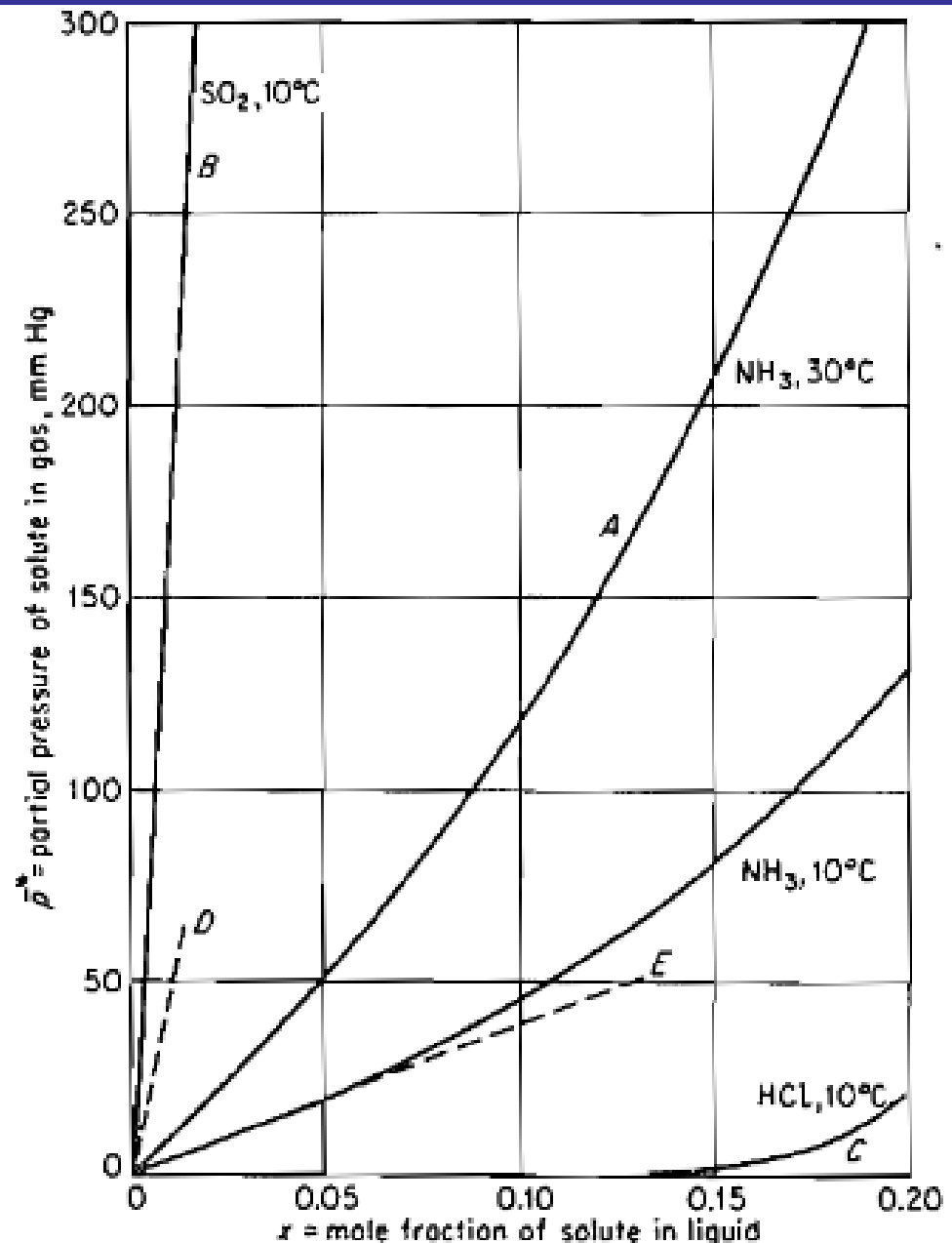
Partial pressure of solute in gas phase (kN/m ²)	Concentration of solute in water kg/1000 kg water		
	Ammonia	Sulphur dioxide	Oxygen
1.3	11	1.9	—
6.7	50	6.8	—
13.3	93	12	0.008
26.7	160	24.4	0.013
66.7	315	56	0.033

- A slightly soluble gas requires a much **higher partial pressure** of the gas in contact with the liquid to give a solution of a given concentration.
- Conversely, with a very soluble gas a given concentration in the liquid phase is obtained with a **lower partial pressure** in the vapour phase

- At 293 K a solution of 4 kg of sulphur dioxide per 1000 kg of water exerts a partial pressure of 2.7 kN/m². If a gas is in contact with this solution with a partial pressure SO₂ greater than 2.7 kN/m², sulphur dioxide will be absorbed.
- The most concentrated solution that can be obtained is that in which the partial pressure of the solute gas is equal to its partial pressure in the gas phase.
- These equilibrium conditions fix the limits of operation of an **absorption unit**.
- The solubility of a gas is not substantially affected by the total pressure in the system for pressures up to about 500 kN/m², it is important to note that the solubility falls with a rise of temperature.

Thus, for a concentration of **10 mole % of ammonia** in water, the equilibrium partial pressure of the ammonia is around 47 kN/m² at 10 °C and around 117 kN/m² at 30 °C

Figure: Solubility of gases in water



- In many instances the absorption is accompanied by the **evolution of heat**, and it is therefore necessary to **fit coolers** to the equipment to **keep the temperature sufficiently low** for an adequate degree of absorption to be obtained
- *Different gases* and *liquids yield* separate ***solubility curves***, which must be determined ***experimentally*** for each system.
- In most cases, the ***solubility of a gas decreases with increasing T***.

- For dilute concentrations of most gases, and over a wide range for some gases, the equilibrium relationship is given by Henry's law.
- This law can be written as:

$$P_A = HC_A \dots\dots\dots (12.1)$$

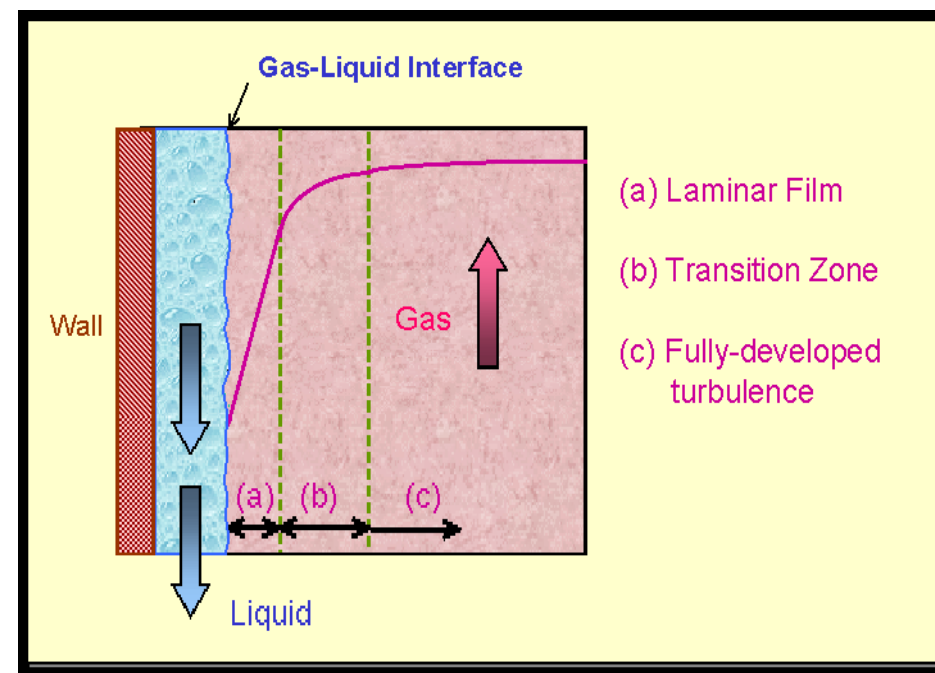
where: P_A is the partial pressure of the component A in the gas phase, C_A is the concentration of the component in the liquid, and H is Henry's constant.

- Absorption data is typically available in the form of solute mole fractions, y_i vs. x_i , or in terms of the Henry's constant, H .
- Henry's Law, in terms of the mole fractions of solute i and the total pressure, is:

$$y_i = \frac{H_i}{P_{\text{Tot}}} x_i \qquad \frac{H_i}{P_{\text{Tot}}} = \frac{y_i}{x_i}$$

The mechanism of absorption

- Gas absorption operation involves mass transfer from the gas phase to the liquid phase. That means the gas molecules must **diffuse from the main body of the gas phase to the gas-liquid interface**, then **cross this interface** into the liquid side, and finally **diffuses from the interface into the main body of the liquid**.



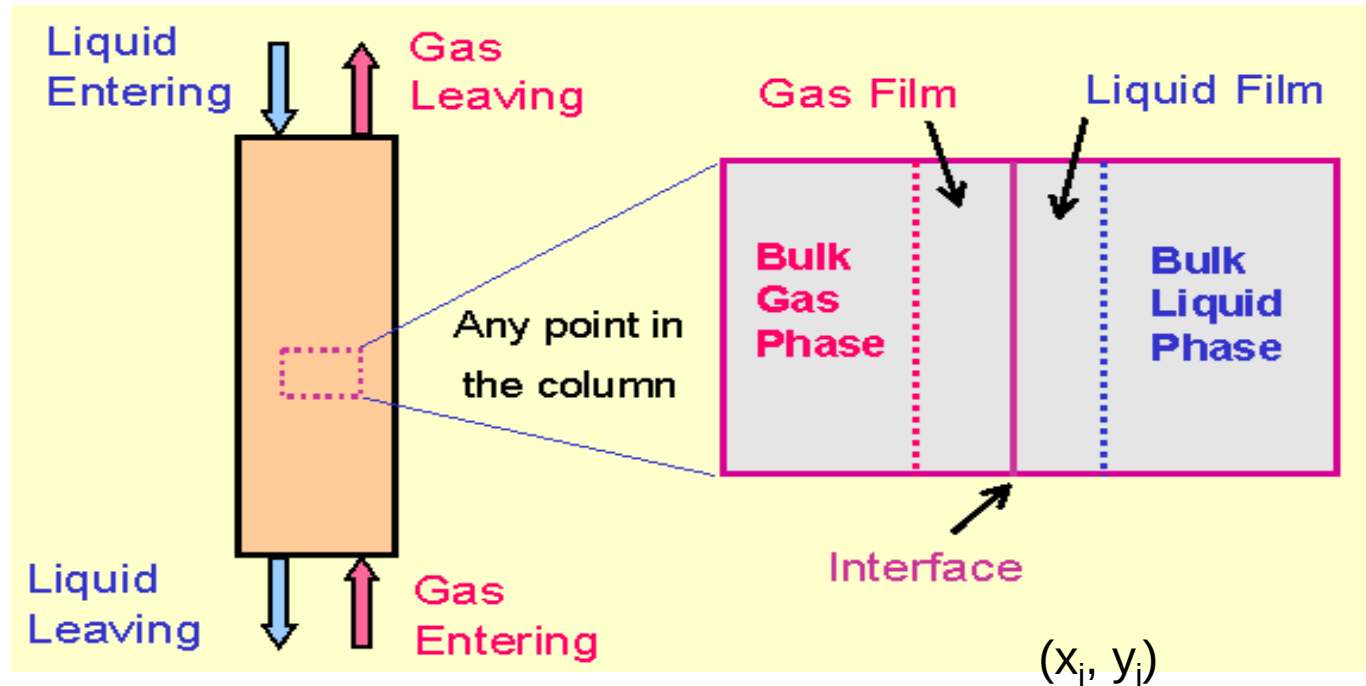
In the gas phase, 3 flow regimes can be visualized :

- A laminar film with molecular diffusion
- A transition zone with some turbulence
- Fully developed turbulent region where most of the mass transfer takes place by eddy diffusion

Two film theory of mass transfer

Assumptions of two-film theory:

- **Steady-state:** concentrations at any position in the tower **do not** change with time.
- Interface between the gas phase and the liquid phase is a **sharp boundary**.
- **Laminar film** exist at the interface on both sides of the interface
- **Equilibrium** exists at the interface, thus there is **negligible resistance** to mass transfer across the interface: (x_i, y_i) is the equilibrium concentration.
- **No chemical reaction:** rate of diffusion across the gas-phase film must equal the rate of diffusion across the liquid-phase film



- According to this theory materials transferred in the **bulk of the phase by convection mechanism** and concentration difference are regarded as negligible.
- Diffusion mass transfer is occurred in area of the **interphase** b/n phases.
- Diffusion of soluble gas **A** in to a liquid, **A** should be diffuse in to a stagnant gas film and then through a stagnant liquid film before entering to the main bulk of the liquid.

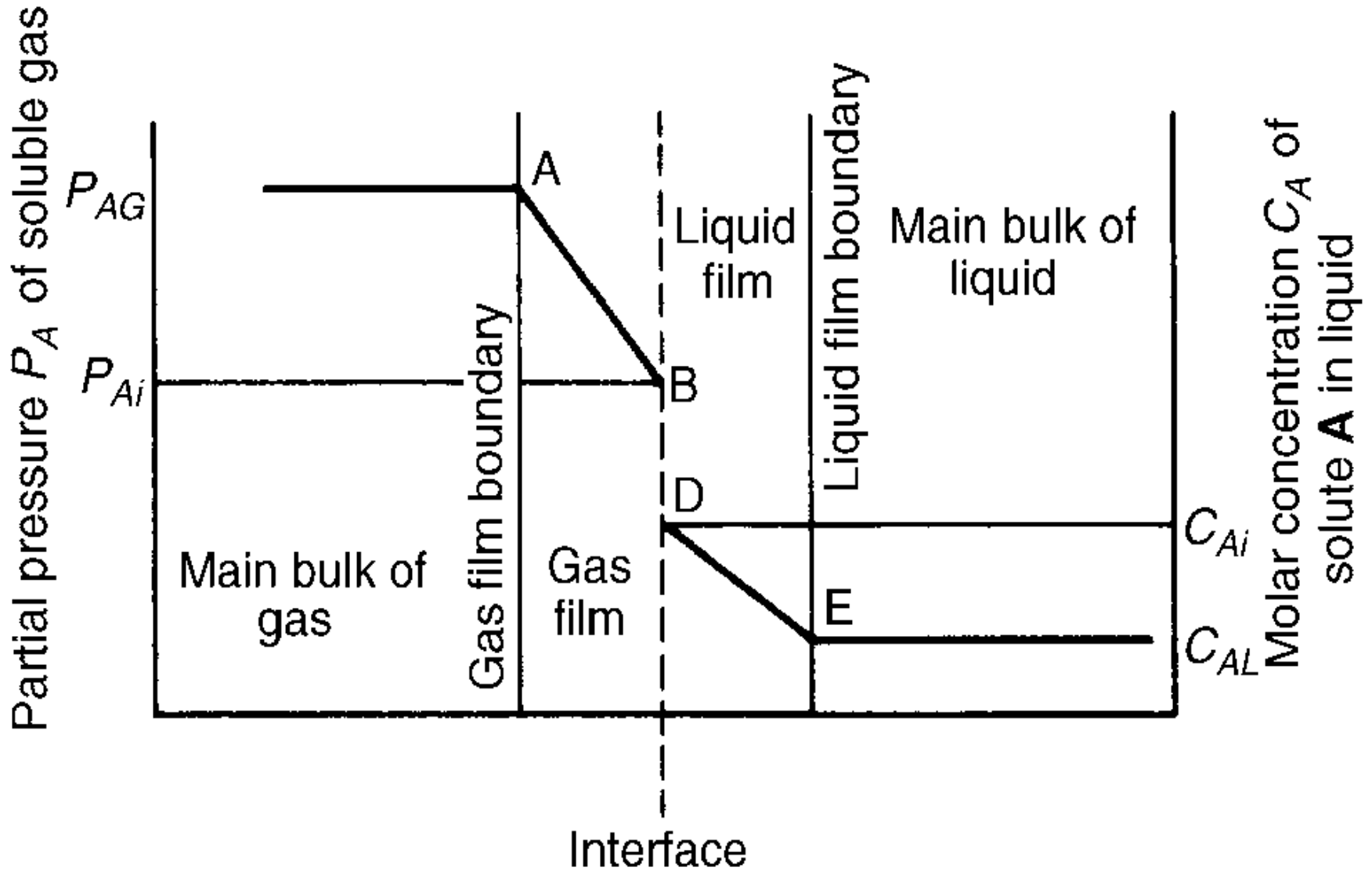


Figure 12.1. Concentration profile for absorbed component **A**

- There is a very large concentration **gradient across the interface**, although this is not the controlling factor in the mass transfer, as it is generally assumed that there is **no resistance at the interface itself**, where equilibrium conditions will exist.
- The controlling factor will be the **rate of diffusion through the two films** where all the resistance is considered to lie.

Diffusion through a stagnant gas

The absorption of a gas consisting of a soluble component A and an insoluble component B is a problem of mass transfer through a stationary gas to which Stefan's law applies

$$N'_A = -D_V \frac{C_T}{C_B} \frac{dC_A}{dz} \quad (12.2)$$

where N'_A is the overall rate of mass transfer (moles/unit area and unit time),
 D_V is the gas-phase diffusivity,
 z is distance in the direction of mass transfer, and
 C_A , C_B , and C_T are the molar concentrations of A, B, and total gas, respectively.

Integrating over the whole thickness z_G of the film, and representing concentrations at each side of the interface by suffixes 1 and 2:

$$N'_A = \frac{D_V C_T}{z_G} \ln \frac{C_{B2}}{C_{B1}} \quad (12.3)$$

Since $C_T = P/RT$, where R is the gas constant, T the absolute temperature, and P the total pressure. For an ideal gas, then:

$$N'_A = \frac{D_V P}{RT z_G} \ln \frac{P_{B2}}{P_{B1}} \quad (12.4)$$

Writing P_{Bm} as the log mean of the partial pressures P_{B1} and P_{B2} , then:

$$P_{Bm} = \frac{P_{B2} - P_{B1}}{\ln(P_{B2}/P_{B1})} \quad (12.5)$$

$$\begin{aligned} N'_A &= \frac{D_V P}{RT z_G} \frac{P_{B2} - P_{B1}}{P_{Bm}} \\ &= \frac{D_V P}{RT z_G} \left[\frac{P_{A1} - P_{A2}}{P_{Bm}} \right] \end{aligned} \quad (12.6)$$

Hence the rate of absorption of A per unit time per unit area is given by:

$$N'_A = k'_G P \left[\frac{P_{A1} - P_{A2}}{P_{Bm}} \right] \quad (12.7)$$

or:
$$N'_A = k_G (P_{A1} - P_{A2}) \quad (12.8)$$

where:
$$k'_G = \frac{D_V}{RT z_G}, \quad \text{and} \quad k_G = \frac{D_V P}{RT z_G P_{Bm}} = \frac{k'_G P}{P_{Bm}} \quad (12.9)$$

- In the great majority of industrial processes the film thickness is not known, so that the rate equation of immediate use is equation 12.8 using k_G .
- k_G is known as the gas-film transfer coefficient for absorption and is a direct measure of the rate of absorption per unit area of interface with a driving force of unit partial pressure difference.

Diffusion in the liquid phase

- The rate of diffusion in liquids is **much slower** than in gases, and mixtures of liquids may take a long time to reach equilibrium unless agitated.
- This is partly due to the **much closer spacing** of the molecules, as a result of which the molecular attractions are more important.
- The basic equation is taken as similar to that for gases, or for dilute concentrations:

$$N'_A = -D_L \frac{dC_A}{dz} \quad (12.10)$$

On integration:

$$N'_A = -D_L \left[\frac{C_{A2} - C_{A1}}{z_L} \right] \quad (12.11)$$

where: C_A, C_B are the molar concentrations of **A** and **B**,
 z_L is the thickness of liquid film through which diffusion occurs, and
 D_L is the diffusivity in the liquid phase.

Since the film thickness is rarely known, equation 12.11 is usually rewritten as:

$$N'_A = k_L (C_{A1} - C_{A2}) \quad (12.12)$$

which is similar to equation 12.8 for gases.

k_L is the liquid-film transfer coefficient, which is usually expressed in $\text{kmol/s m}^2(\text{kmol/m}^3) = \text{m/s}$. For dilute concentrations:

$$k_L = D_L/z_L$$

Rate of absorption

In a **steady-state process** of absorption, the rate of transfer of material through the gas film will be the same as that through the liquid film, and the general equation for mass transfer of a component A may be written as:

$$N'_A = k_G(P_{AG} - P_{Ai}) = k_L(C_{Ai} - C_{AL}) \quad (12.13)$$

where P_{AG} is the partial pressure in the bulk of the gas, C_{AL} is the concentration in the bulk of the liquid, and P_{Ai} and C_{Ai} are the values of concentration at the interface where equilibrium conditions are assumed to exist. Therefore:

$$\frac{k_G}{k_L} = \frac{C_{Ai} - C_{AL}}{P_{AG} - P_{Ai}} \quad (12.14)$$

These conditions may be illustrated graphically as in Figure 12.2, where ABF is the equilibrium curve for the soluble component A.

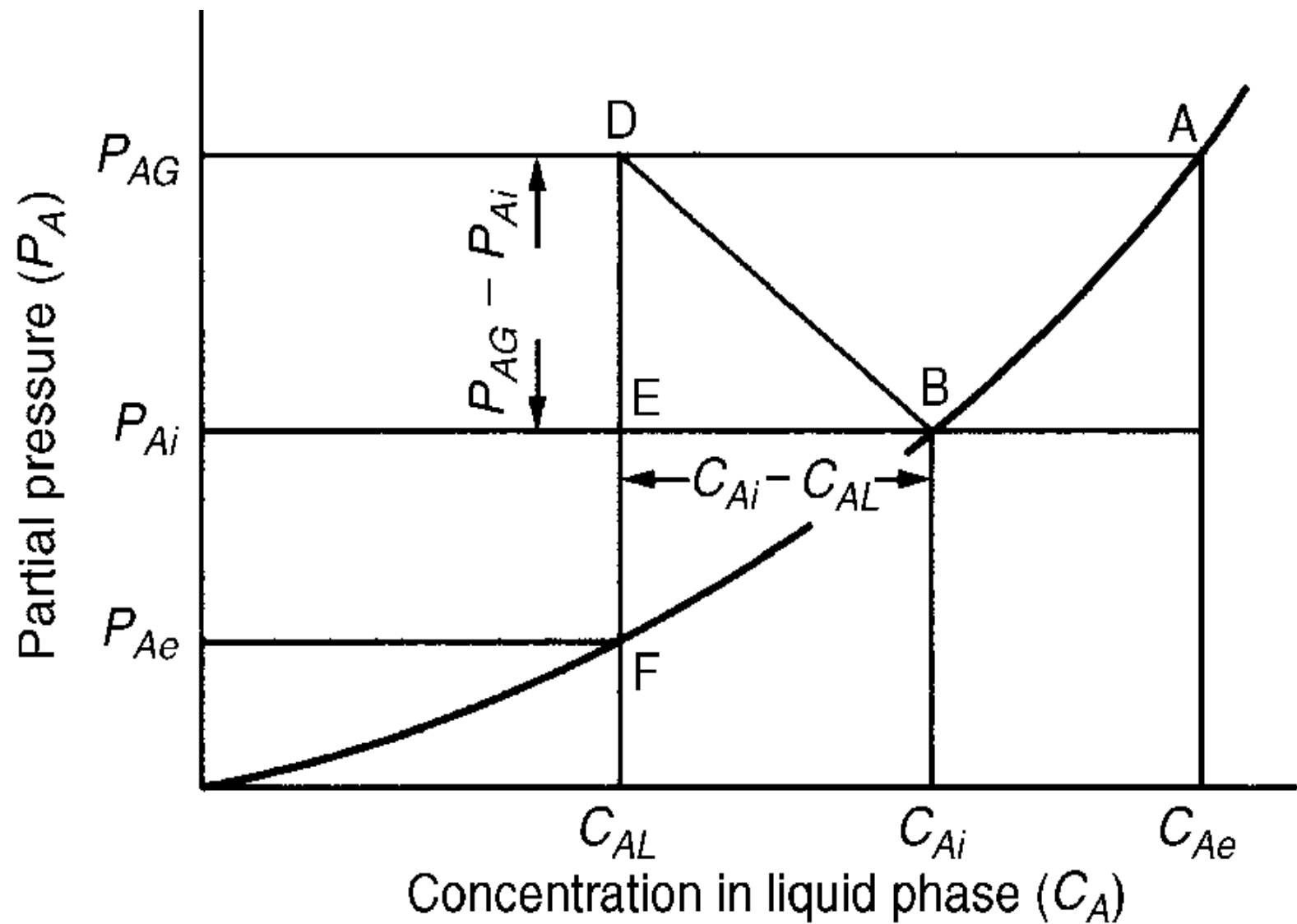


Figure 12.2. Driving forces in the gas and liquid phases

Point D (C_{AL} , P_{AG}) represents conditions in the bulk of the gas & liquid.

Point A (C_{Ae} , P_{AG}) represents a concentration of C_{Ae} in the liquid in equilibrium with P_{AG} in the gas.

Point B (C_{Ai} , P_{Ai}) represents the concentration of C_{Ai} in the liquid in equilibrium with P_{Ai} in the gas, and gives conditions at the interface.

Point F (C_{AL} , P_{Ae}) represents a partial pressure P_{Ae} in the gas phase in equilibrium with C_{AL} in the liquid.

Then, the driving force causing transfer in the gas phase is:

$(P_{AG} - P_{Ai}) \equiv DE$ and the driving force causing transfer in the liquid phase is: $(C_{Ai} - C_{AL}) \equiv BE$ Then:

$$(P_{AG} - P_{Ai}) / (C_{Ai} - C_{AL}) = k_L / k_G$$

and the concentrations at the interface (point B) are found by drawing a line through D of slope $-k_L/k_G$ to cut the equilibrium curve in B.

Overall coefficients

- In order to obtain a direct measurement of the values of k_L and k_G the **measurement of the concentration at the interface** would be necessary.
- These values can only be obtained in very special circumstances, and it has been found of considerable value to use two overall coefficients K_G and K_L defined by:

$$N'_A = K_G(P_{AG} - P_{Ae}) = K_L(C_{Ae} - C_{AL}) \quad (12.15)$$

K_G and K_L are known as the overall gas and liquid phase coefficients, respectively.

Relation between film and overall coefficients

The rate of transfer of **A** may now be written as:

$$N'_A = k_G[P_{AG} - P_{Ai}] = k_L[C_{Ai} - C_{AL}] = K_G[P_{AG} - P_{Ae}] = K_L[C_{Ae} - C_{AL}]$$

Thus:

$$\begin{aligned} \frac{1}{K_G} &= \frac{1}{k_G} \left[\frac{P_{AG} - P_{Ae}}{P_{AG} - P_{Ai}} \right] \\ &= \frac{1}{k_G} \left[\frac{P_{AG} - P_{Ai}}{P_{AG} - P_{Ai}} \right] + \frac{1}{k_G} \left[\frac{P_{Ai} - P_{Ae}}{P_{AG} - P_{Ai}} \right] \end{aligned} \quad (12.16)$$

From the previous discussion:

$$\frac{1}{k_G} = \frac{1}{k_L} \left[\frac{P_{AG} - P_{Ai}}{C_{Ai} - C_{AL}} \right]$$

Thus:

$$\begin{aligned}\frac{1}{K_G} &= \frac{1}{k_G} + \frac{1}{k_L} \left[\frac{P_{AG} - P_{Ai}}{C_{Ai} - C_{AL}} \right] \left[\frac{P_{Ai} - P_{Ae}}{P_{AG} - P_{Ai}} \right] \\ &= \frac{1}{k_G} + \frac{1}{k_L} \left[\frac{P_{Ai} - P_{Ae}}{C_{Ai} - C_{AL}} \right]\end{aligned}$$

$(P_{Ai} - P_{Ae})/(C_{Ai} - C_{AL})$ is the average slope of the equilibrium curve and, when the solution obeys Henry's law, $\mathcal{H} = dP_A/dC_A \approx (P_{Ai} - P_{Ae})/(C_{Ai} - C_{AL})$.

Therefore:

$$\frac{1}{K_G} = \frac{1}{k_G} + \frac{\mathcal{H}}{k_L} \quad (12.17)$$

Similarly:

$$\frac{1}{K_L} = \frac{1}{k_L} + \frac{1}{\mathcal{H}k_G} \quad (12.18)$$

and:

$$\frac{1}{K_G} = \frac{\mathcal{H}}{K_L} \quad (12.19)$$

The equilibrium constant $\mathcal{H}$ must be constant, there must be **no significant interfacial resistance**, and there must be no interdependence of the values of the two film-coefficients.

Example 12.1

The overall liquid transfer coefficient, $K_L a$, for the absorption of SO_2 in water in a column is $0.003 \text{ kmol/s m}^3 \text{ (kmol/m}^3\text{)}$. By assuming an expression for the absorption of NH_3 in water at the same liquor rate and varying gas rates, derive an expression for the overall liquid film coefficient $K_L a$ for absorption of NH_3 in water in this equipment at the same water rate though with varying gas rates. The diffusivities of SO_2 and NH_3 in air at 273 K are 0.103 and $0.170 \text{ cm}^2/\text{s}$. SO_2 dissolves in water, and Henry's constant is equal to $50 \text{ (kN/m}^2\text{)/(kmol/m}^3\text{)}$. All data are expressed for the same temperature.

Solution

From equation 12.18:

$$\frac{1}{K_L a} = \frac{1}{k_L a} + \frac{1}{\mathcal{H} k_G a} = \frac{1}{0.003} = 333.3$$

For the absorption of a moderately soluble gas it is reasonable to assume that the liquid and gas phase resistances are of the same order of magnitude, assuming them to be equal.

$$\frac{1}{k_L a} = \frac{1}{\mathcal{H} k_G a} = \left(\frac{333}{2} \right) = 166.7$$

or: $k_L a = \mathcal{H} k_G a = 0.006 \text{ kmol/s m}^3 \text{ (kmol/m}^3\text{)}$

Thus, for SO_2 : $k_G a = 0.006/\mathcal{H} = 0.006/50 = 0.00012 \text{ kmol/s m}^3 \text{ (kN/m}^2\text{)}$

From equation 12.25: $k_G a$ is proportional to $(\text{diffusivity})^{0.56}$.

Hence for NH_3 : $k_G a = 0.00012(0.17/0.103)^{0.56} = 0.00016 \text{ kmol/s m}^3 \text{ (kN/m}^2\text{)}$

$$\frac{h_D}{u} \frac{P_{Bm}}{P} \left[\frac{\mu}{\rho D_V} \right]^{0.56}$$

For a very soluble gas such as NH_3 , $k_G a \simeq K_G a$.

For NH_3 the liquid-film resistance will be small, and:

$$k_G a = K_G a = \underline{\underline{0.00016 \text{ kmol/s m}^3 \text{ (kN/m}^2\text{)}}}$$

Packed Columns

- Packed columns are used for distillation, gas absorption, and liquid-liquid extraction.
- The gas liquid contact in a packed bed column is continuous, not stage-wise, as in a plate column.
- The liquid flows down the column over the packing surface and the gas or vapor, counter-currently, up the column. Some gas-absorption columns are co-current
- The performance of a packed column is very dependent on the maintenance of good liquid and gas distribution throughout the packed bed.

- It consists of a **cylindrical column or tower** equipped with a
 - ✓ gas **inlet** and **distributing space** at the **bottom**
 - ✓ **liquid distributor** at the top
 - ✓ gas and liquid **outlets** at the **top** and **bottom**, respectively and
 - ✓ a supported mass of **inert solid shapes** called **tower packing**.
- The **packing support** is typically a **screen**, corrugated to give **strength**, with a large **open area** so that **flooding does not** occur at the support.
- The **inlet liquid** which may be **pure solvent** or a **dilute solution** of solute in the solvent and which is called the **weak liquor** is distributed over the top of packing by the **distributor**

- In *ideal operation*, uniformly wets the surfaces of the packing.
- The **distributor** is a *set of perforated pipes*
- In *large towers* *spray nozzles or distributor* plates with *overflow weirs* are more common
- The *solute containing gas* (rich gas) enters the *distributing space below the packing* and *flows upward* through the *interstices* in the *packing counter current* to the *flow of the liquid*.
- The **packing** provides a *large area of contact* between the *liquid* and *gas* and encourages *intimate contact between the phases*.
- The *solute* in the *rich gas* is absorbed by the *fresh liquid* entering the tower and *dilute or lean gas* leaves the top.

- The ***liquid is enriched in solute*** as it flows down the tower and ***concentrated liquid*** called ***strong liquor***, leaves the bottom of the tower through the *liquid outlet*.
- ***Tower packing*** are divided into two ***principal types***. These are:
 - ✓ ***Dumped*** at ***random*** into tower
 - ✓ ***Structured or ordered*** packing
- ***Dumped packing*** consists of units ***6 to 75 mm*** in major dimension; ***packing smaller*** than 25 mm are used mainly in ***laboratory or pilot plant columns***
- In ***stacked packing*** the units are ***50 to 200 mm*** in sizes. They are ***less commonly*** used than ***dumped packing***.

Dumped tower packing

- Are made of *cheap, inert materials* such as *clay, porcelain or various plastics*
- *Thin walled metals rings* of steel or aluminium are sometimes used
- *High void spaces* and *large passages* for the fluids are achieved by making the packing *units irregular or hollow*, so that they interlock into open structures with a *porosity or void fraction* of *60 – 90%*

Structured packings

- With *ordered geometry*
- *Early structured packing* were fabricated from *wire gauze*, most current ones are made of *sheets of perforated corrugated metals*, with adjacent *sheets arranged* so that *liquid spreads* over their surfaces while flows through *channels* formed by *corrugations*.

Contact between Liquid and Gas

- Good contact between *liquid* and *gas* is the hardest to meet
- Ideally, the **liquid**, once *distributed over the top of the packing*, flows in *thin films* over ***all the packing surfaces*** all the way ***down the tower***
- Actually, the ***films tend to grow thicker*** in some places and *thinner in others*, so that the liquid collects into *small rivulets* and *flows along* localized paths through the ***packing***
- Especially at ***low liquid rates*** much of the packing surface may be *dry or at best*, covered by a ***stagnant film of liquid***
- Three main problems for flow of liquid through a specific packing, all characteristic of flow and always independent of MT itself
 - i. Channeling;
 - ii. Loading;
 - iii. Flooding

Channelling:

- Gas or liquid flow is much greater at some points than at others
- Can substantially reduce MT
- Can be severe in stacked packings
- Usually minor in crushed solid packings
- Minimal in commercial random packings.

Loading

- Imagine a liquid flowing down a packed bed
- A small gas flow introduced at the bottom
- At small gas flow rates, liquid flow unaltered
- As gas flow is increased, liquid flow diminishes at some value of gas flow



Loading!

Flooding

- At very high gas flow rates, the liquid flow stops altogether and liquid collects at the top of the column



- It dramatically reduces MT.
- Flooding is avoided by reducing liquid and gas fluxes.

Pressure drop and limiting flow rates

- The *pressure drop per initial packing depth* comes from the **fluid friction**

$$G_y = u_o \rho_y$$

G_y = mass of gas per hours per unit of cross sectional area

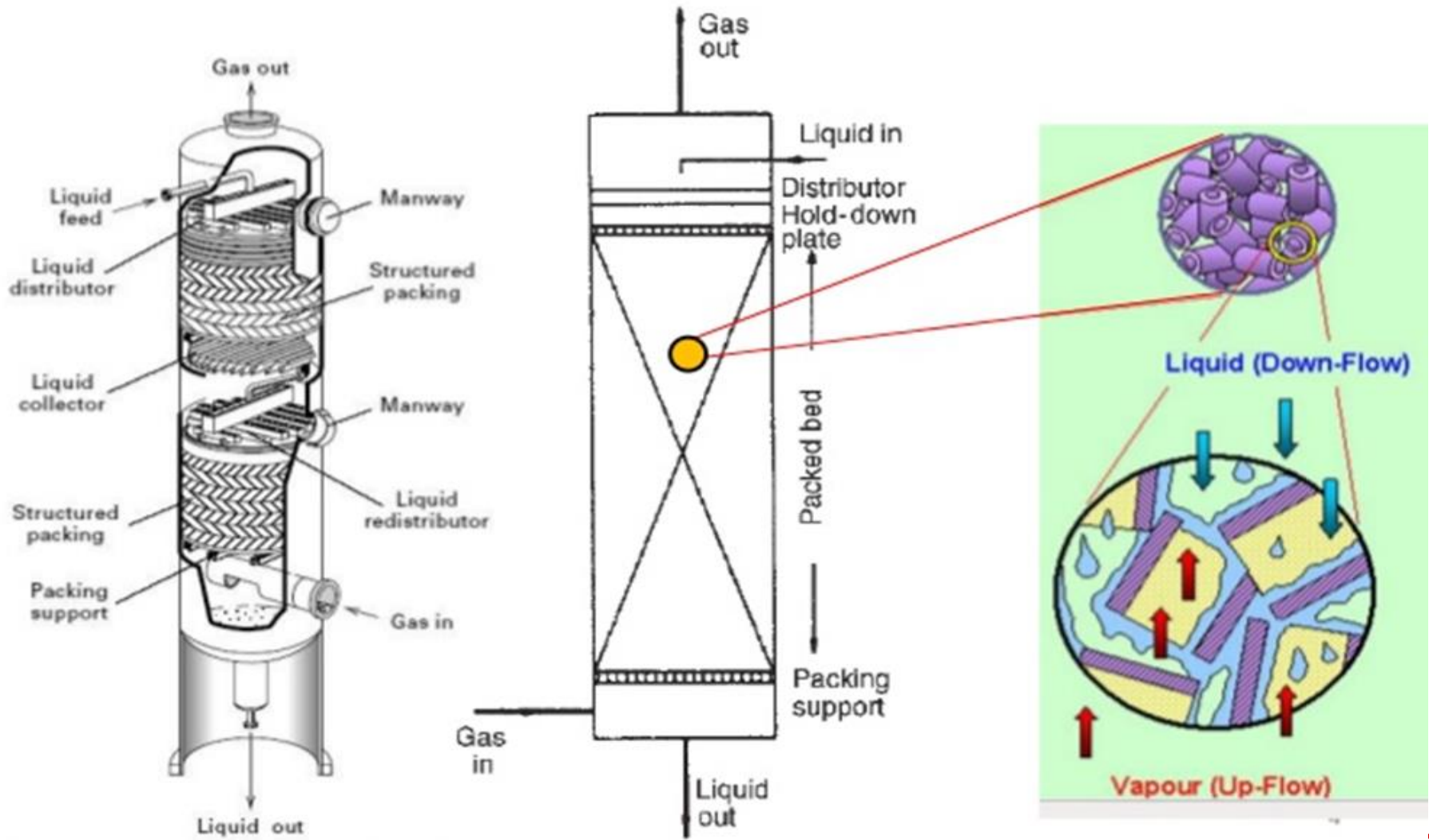
u_o = superficial gas velocity

ρ_y = the density of the gas

- **Higher gas flow** brings up liquid *rapidly accumulates* (entrained) and the entire column may fill with *liquid*.
- The **void fraction**, however, *doesn't change* with **gas flow**.
- The *gas velocity* in an operating **packed column** must obviously be *lower than the flooding velocity*.

- However, as *flooding approached*, *most or all* of the packing *surface is wetted*, *maximizing* the *contact area between gas & the liquid*.
- *Lowering*, the *design gas velocity increases* the *tower diameter* without much change in the required **height**, since *lower gas and liquid velocities* head to a nearly proportional reduction in *mass transfer rate*.
- *Decreased pressure drop* is one benefit of *low gas velocity* but the *cost of power* consumed is usually *not a major factors* in optimizing the design.

Components of a Packed Column



Advantages of Packed Columns

1. For corrosive liquids, a packed column will usually be cheaper than the equivalent plate column.
2. The liquid hold-up is lower in a packed column than a plate column. This can be important when the inventory of toxic or flammable liquids needs to be kept as small as possible for safety reasons.
3. Packed columns are more suitable for handling foaming systems.
4. The pressure drop can be lower for packing than plates; and packing should be considered for vacuum columns.
5. Packing should always be considered for small diameter columns, say less than 0.6 m, where plates would be difficult to install, and expensive.

When to use packed or plate absorption column?

Packed absorbers are preferred

1. When the required **diameter** is **less** than 60 cm
2. The **pressure drop** must be **low**, as for a vacuum service
3. **Corrosion** considerations favour the use of **ceramic or polymeric materials**
4. **Low liquid holdup** is desirable

Plate absorbers are preferred

1. The **liquid/gas** ratio is very low
2. Frequent **cleaning** is required
3. **Large diameter**
4. For **high pressure drop process**
 - If there is no overriding consideration, **cost is the major factor** to be taken into account when packed or tray type absorption or stripping column

Packed Towers For Gas Absorption

- From the analysis of the diffusional nature of absorption, one of the outstanding requirements is to provide **large an interfacial area of contact between the phases.**
- For this purpose, columns similar to those used for distillation are suitable, distillation columns are usually tall and thin but absorption columns are more likely to be **short** and **fat.**
- In addition, equipment may be used in which gas is passed into a liquid which is agitated by a stirrer.
- A few special forms of units have also been used, although it is the packed column which is most frequently used for **gas absorption applications.**

Capacity of packed towers

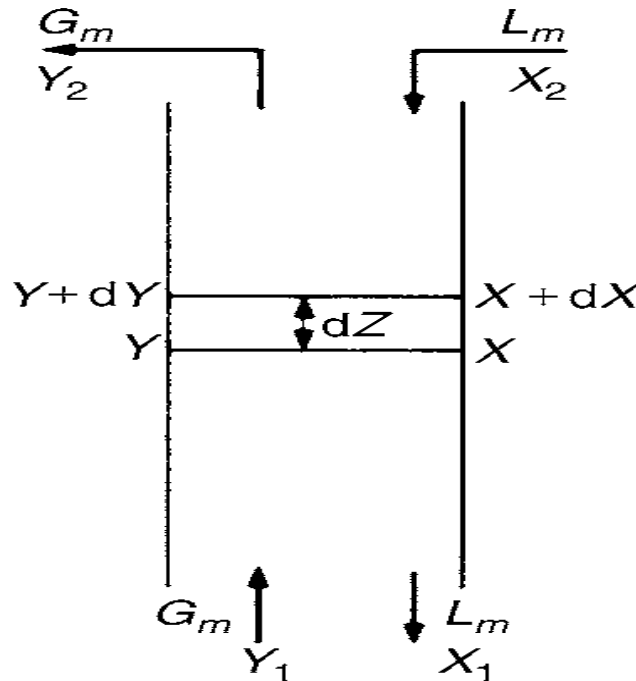


Figure 12.18. Countercurrent absorption tower

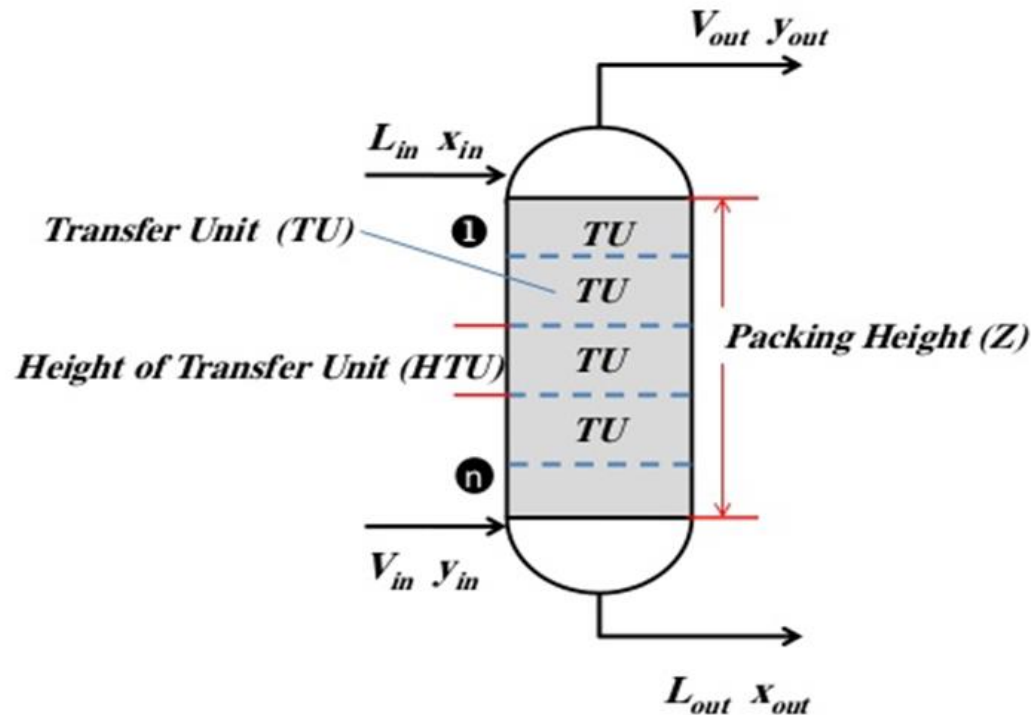
$$\begin{aligned}
 N'_A A dZ a &= k_G a (P_{AG} - P_{Ai}) A dZ \\
 &= k_L a (C_{Ai} - C_{AL}) A dZ
 \end{aligned}
 \tag{12.41}$$

where: N'_A = kmol of solute absorbed per unit time and unit interfacial area,
 a = surface area of interface per unit volume of column,
 A = cross-sectional area of column, and
 Z = height of packed section.

$$\text{The interfacial area for transfer} = a dV = a A dZ
 \tag{12.42}$$

The essential features of a packed column are the **shell**, the **arrangements for the gas and liquid inlets and outlets** and the **packing with its necessary supporting and redistributing systems**. This section is largely concerned with the determination of the **height of packing** for a particular duty.

Packing Height (Z)



$$\text{Packing Height (Z)} = \text{height of transfer unit (HTU)} \times \text{number of transfer units (n)}$$

Height of column based on conditions in the gas film

If G_m = moles of inert gas/(unit time) (unit cross-section of tower),

L_m = moles of solute-free liquor/(unit time) (unit cross-section of tower),

Y = moles of solute gas **A**/mole of inert gas **B** in gas phase, and

X = moles of solute **A**/mole of inert solvent in liquid phase.

and at any plane at which the molar ratios of the diffusing material in the gas and liquid phases are Y and X , then over a small height dZ , the moles of gas leaving the gas phase will equal the moles taken up by the liquid.

Thus:
$$AG_m dY = AL_m dX \quad (12.43)$$

But:
$$G_m A dY = N'_A (a dV) = k_G a (P_{Ai} - P_{AG}) A dZ \quad (12.44)$$

It may be noted that, in a gas absorption process, gas and liquid concentrations will decrease in the upwards direction and both dX and dY will be negative.

Since:
$$P_{AG} = \frac{Y}{1 + Y} P$$

$$\begin{aligned} G_m dY &= k_G a P \left[\frac{Y_i}{1 + Y_i} - \frac{Y}{1 + Y} \right] dZ \\ &= k_G a P \left[\frac{Y_i - Y}{(1 + Y)(1 + Y_i)} \right] dZ \end{aligned}$$

Hence the height of column Z required to achieve a change in Y from Y_1 at the bottom to Y_2 at the top of the column is given by:

$$\int_0^Z dZ = Z = \frac{G_m}{k_G a P} \int_{Y_1}^{Y_2} \frac{(1+Y)(1+Y_i) dY}{Y_i - Y} \quad (12.45)$$

which for dilute mixtures may be written as:

$$Z = \frac{G_m}{k_G a P} \int_{Y_1}^{Y_2} \frac{dY}{Y_i - Y} \quad (12.46)$$

In this analysis it has been assumed that k_G is a constant throughout the column, and provided the concentration changes are not too large this will be reasonably true.

Height of column based on conditions in liquid film

A similar analysis may be made in terms of the liquid film. Thus from equations 12.41 and 12.42:

$$AL_m dX = k_L a (C_{Ai} - C_{AL}) A dZ \quad (12.47)$$

where the concentrations C are in terms of moles of solute per unit volume of liquor. If $C_T = (\text{moles of solute} + \text{solvent}) / (\text{volume of liquid})$, then:

$$\frac{C_A}{C_T - C_A} = \frac{\text{moles of solute}}{\text{moles of solvent}} = X$$

whence:

$$C_A = \frac{X}{1+X} C_T \quad (12.48)$$

The transfer equation (12.47) may now be written as:

$$\begin{aligned} L_m dX &= k_L a C_T \left[\frac{X}{1+X} - \frac{X_i}{1+X_i} \right] dZ \\ &= k_L a C_T \left[\frac{X - X_i}{(1+X_i)(1+X)} \right] dZ \end{aligned}$$

Thus:
$$\int_0^Z dZ = Z = \frac{L_m}{k_L a C_T} \int_{X_1}^{X_2} \frac{(1+X_i)(1+X) dX}{X - X_i} \quad (12.49)$$

and for dilute concentrations this gives:

$$Z = \frac{L_m}{k_L a C_T} \int_{X_1}^{X_2} \frac{dX}{X - X_i} \quad (12.50)$$

where C_T and k_L have been taken as constant over the column.

Height based on overall coefficients

If the driving force based on the gas concentration is written as $(Y - Y_e)$ and the overall gas transfer coefficient as K_G , then the height of the tower for dilute concentrations becomes:

$$Z = \frac{G_m}{K_G a P} \int_{Y_1}^{Y_2} \frac{dY}{Y_e - Y} \quad (12.51)$$

or in terms of the liquor concentration as:

$$Z = \frac{L_m}{K_L a C_T} \int_{X_1}^{X_2} \frac{dX}{X - X_e} \quad (12.52)$$

Equations for dilute concentrations

As the mole fraction is approximately equal to the molar ratio at dilute concentrations then considering the gas film:

$$Z = \frac{G_m}{K_G a P} \int_{Y_1}^{Y_2} \frac{dY}{Y_e - Y} = \frac{G_m}{K_G a P} \int_{y_1}^{y_2} \frac{dy}{y_e - y} \quad (12.53)$$

and considering the liquid film:

$$Z = \frac{L_m}{K_L a C_T} \int_{X_1}^{X_2} \frac{dX}{X - X_e} = \frac{L_m}{K_L a C_T} \int_{x_1}^{x_2} \frac{dx}{x - x_e} \quad (12.54)$$

The operating line and graphical integration for the height of a column

Taking a material balance on the solute from the bottom of the column to any plane where the mole ratios are Y and X gives for unit area of cross-section:

$$G_m(Y_1 - Y) = L_m(X_1 - X) \quad (12.55)$$

or:
$$Y_1 - Y = \frac{L_m}{G_m}(X_1 - X) \quad (12.56)$$

This is the equation of a straight line of slope L_m/G_m , which passes through the point (X_1, Y_1) . It may be seen by making a material balance over the whole column that the same line passes through the point (X_2, Y_2) . This line, known as the operating line,

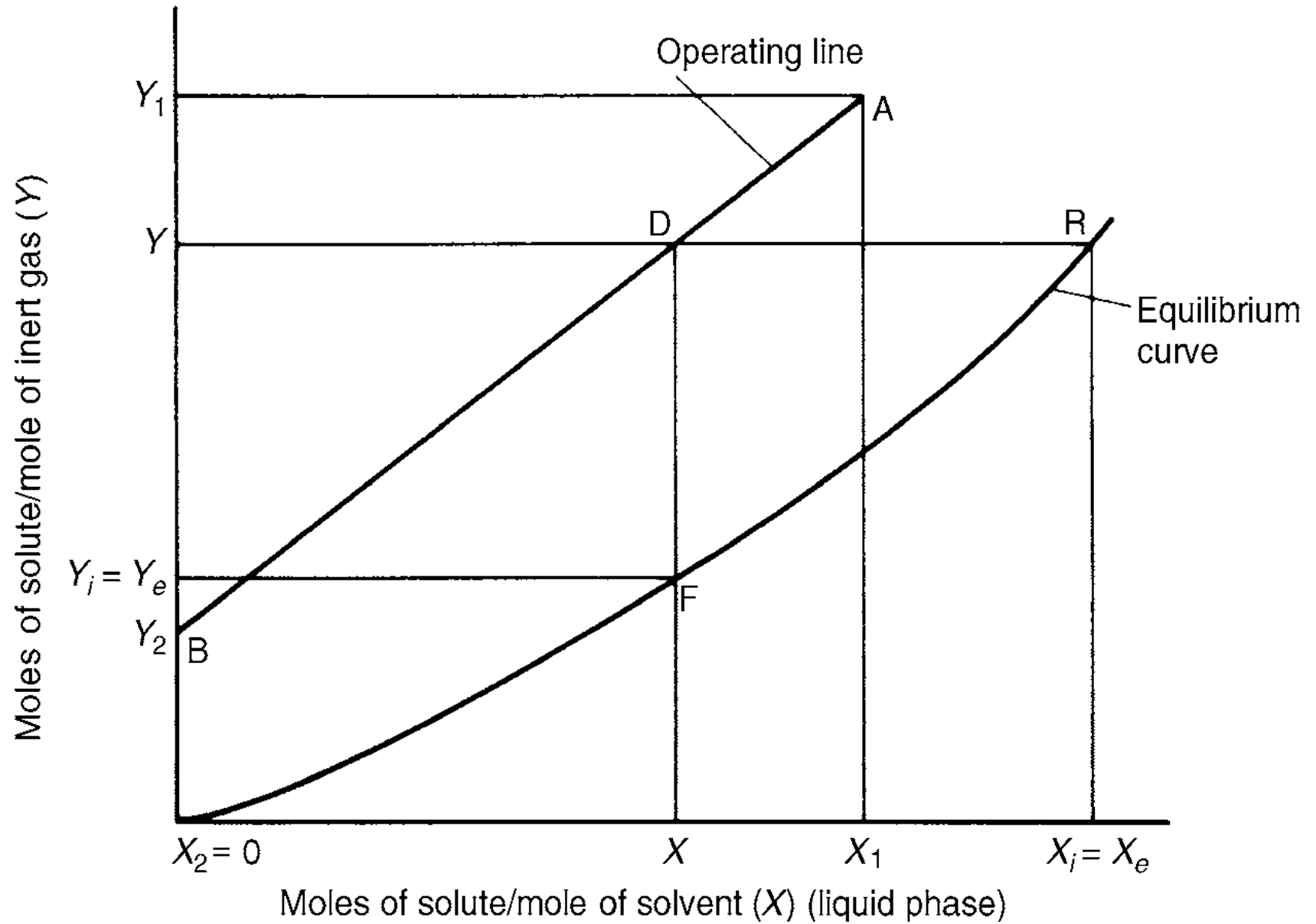


Figure 12.19. Driving force in gas and liquid-film controlled processes.

- Figure 12.19 illustrates typical conditions for the case of moist air and sulphuric acid or caustic soda solution, where the main resistance lies in the gas phase.
- The equilibrium curve is represented by the **line FR**, and the operating line is given by **AB**, A corresponding to the concentrations at the bottom of the column and B to those at the top of the column.
- D represents the condition of the bulk of the liquid and gas at any point in the column, and has coordinates X and Y .
- *Then, if the gas film is controlling the process, Y_i equals Y_e , and is given by a point F on the equilibrium curve, with coordinates X and Y_i*

- The driving force causing transfer is then given by the distance **DF**.
- It is therefore possible to evaluate the expression

$$\int_{Y_1}^{Y_2} \frac{dY}{Y_i - Y}$$

by selecting values of Y , reading off from the Figure the corresponding values of Y_i , and thus calculating $1/(Y_i - Y)$. It may be noted that, for gas absorption, $Y > Y_i$ and $Y_i - Y$ and dY in the integral are both negative.

If the liquid film controls the process, X_i equals X_e and the driving force $X_i - X$ is given in Figure 12.19 by the line **DR**. The evaluation of the integral:

$$\int_{X_1}^{X_2} \frac{dX}{X - X_i}$$

may be effected in the same way as for the gas film.

The transfer unit

- The group $\int \frac{dy}{(y_e - y)}$ is defined as the number of overall gas transfer units N_{OG} .
- Over a small height dZ , the partial pressure of the diffusing **component A** will change by an amount dP_{AG} .
- Then the moles of A transferred are given by:

(change in mole fraction) $\times$ (total moles of gas)

Therefore:
$$K_G a (P_{AG} - P_{Ae}) dZ = \frac{-dP_{AG}}{P} G'_m \quad (12.69)$$

(for dilute concentrations)

Thus:
$$\int_{P_{AG_2}}^{P_{AG_1}} \frac{dP_{AG}}{P_{Ae} - P_{AG}} = \int_0^Z \frac{K_G a P}{G'_m} dZ \quad (12.70)$$

or in terms of mole fractions:

$$N_{OG} = \int_{y_1}^{y_2} \frac{dy}{y_e - y} = \int_0^Z K_G a \frac{P}{G'_m} dZ = K_G a \frac{P}{G'_m} Z \quad (12.71)$$

$$N_{OG} = \frac{\text{Height of column}}{\text{Height of transfer unit}} = \frac{Z}{H_{OG}} \quad (12.73)$$

The height of the overall gas transfer unit is then $H_{OG} = \frac{G'_m}{P K_G a}$ (12.74)

If the driving force is taken over the gas-film only, the height of a gas-film transfer unit $H_G = G'_m / P k_G a$ is obtained. Similarly for the liquid film, the height of the overall liquid-phase transfer unit H_{OL} is given by:

$$H_{OL} = \frac{L'_m}{K_L a C_T} \quad (12.75)$$

The height of the liquid-film transfer unit is given by:

$$H_L = \frac{L'_m}{k_L a C_T} \quad (12.76)$$

where C_T is the mean molar density of the liquid.

➤ *Reading assignment: show that*

$$a A Z N'_A = G_m (y_1 - y_2) A = K_G a A P (y - y_e)_{\ln} Z \quad (12.62)$$

When a special case of equilibrium curve is a straight line

Example 12.4

Gas, from a petroleum distillation column, has a concentration of H_2S reduced from 0.03 (kmol H_2S /kmol of inert hydrocarbon gas) to 1 per cent of this value by scrubbing with a triethanolamine–water solvent in a countercurrent tower, operating at 300 K and atmospheric pressure. The equilibrium relation for the solution may be taken as $Y_e = 2X$.

The solvent enters the tower free of H_2S and leaves containing 0.013 kmol of H_2S /kmol of solvent. If the flow of inert gas is 0.015 kmol/s m^2 of tower cross-section, calculate:

- (a) the height of the absorber necessary, and
- (b) the number of transfer units N_{OG} required.

The overall coefficient for absorption $K_G'' a$ may be taken as 0.04 kmol/s m^3 (unit mole fraction driving force).

Solution

$$\text{Driving force at top of column} = (Y_2 - Y_{2e}) = 0.0003$$

$$\text{Driving force at bottom of column} = (Y_1 - Y_{1e}) = (0.03 - 0.026) = 0.004$$

$$\begin{aligned}\text{Logarithmic mean driving force} &= \frac{(0.004 - 0.0003)}{\ln\left(\frac{0.004}{0.0003}\right)} \\ &= 0.00143\end{aligned}$$

From equation 12.62:

$$\text{That is:} \quad G'_m(Y_1 - Y_2) = K''_G a(Y - Y_e)_{\text{lm}} Z$$

$$\text{Thus:} \quad 0.015(0.03 - 0.0003) = 0.04 \times 0.00143 Z$$

$$\text{and:} \quad Z = \frac{0.000446}{0.0000572} = 7.79 = 7.8 \text{ m (say)}$$

$$\begin{aligned}\text{Height of transfer unit } \mathbf{H}_{OG} &= \frac{G'_m}{K''_G a} \\ &= \frac{0.015}{0.04} = 0.375 \text{ m}\end{aligned}$$

$$\text{Number of transfer units } \mathbf{N}_{OG} = \frac{7.79}{0.375} = 20.7 = \underline{\underline{21 \text{ (say)}}}$$

Example 12.5

Ammonia is to be removed from a 10 per cent ammonia–air mixture by countercurrent scrubbing with water in a packed tower at 293 K so that 99 per cent of the ammonia is removed when working at a total pressure of 101.3 kN/m².

If the gas rate is 0.95 kg/m²s of tower cross-section and the liquid rate is 0.65 kg/m²s, find the necessary height of the tower if the absorption coefficient $K_G a = 0.001 \text{ kmol/m}^3\text{s(kN/m}^2\text{)}$ partial pressure difference. The equilibrium data are:

kmol NH ₃ /kmol water:	0.021	0.031	0.042	0.053	0.079	0.106	0.159
Partial pressure NH ₃ :							
(mm Hg)	12.0	18.2	24.9	31.7	50.0	69.6	114.0
(kN/m ²)	1.6	2.4	3.3	4.2	6.7	9.3	15.2

See Coulson Volume 2. page 700 for the solution

Plate/tray Columns For Gas Absorption

- The *main requirement* of a *tray* is that it should provide *intimate mixing* between the *liquid and vapour streams*
- The *arrangements* for the *liquid flow* over the *tray* depend largely on the *ratio of liquid to vapor flow*.
- As *liquid reflex* flows *across each tray* and enters the *down comer* by way of a *weir*, the height of which largely determines the amount of *liquid on the tray*

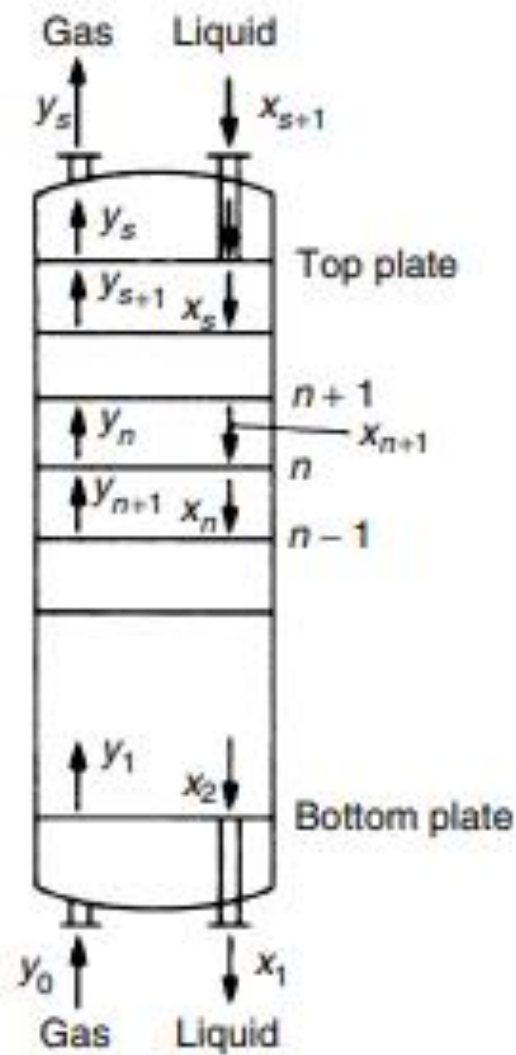


Figure 12.26. Plate tower—nomenclature for fluid streams

Graphical method to determine number of plates

A material balance for the absorbed component from the bottom to a plane above plate n gives

$$G'_m y_n + L'_m x_1 = G'_m y_0 + L'_m x_{n+1} \quad (12.84)$$

$$y_n = \frac{L'_m}{G'_m} x_{n+1} + y_0 - \frac{L'_m}{G'_m} x_1 \quad (12.85)$$

L'_m is the molar rate of flow per unit area of solute free liquid,

G'_m is the molar rate of flow per unit area of inert gas,

n refers to the plate numbered from the bottom upwards (and suffix n refers to material leaving plate n),

x is the mole fraction of the absorbed component in the liquid,

y is the mole fraction of the absorbed component in the gas, and

s is the total number of plates in the column.

This is the equation of a straight line of slope L'_m/G'_m , relating the composition of the gas entering a plate to the liquid leaving the plate, and is known as the operating line

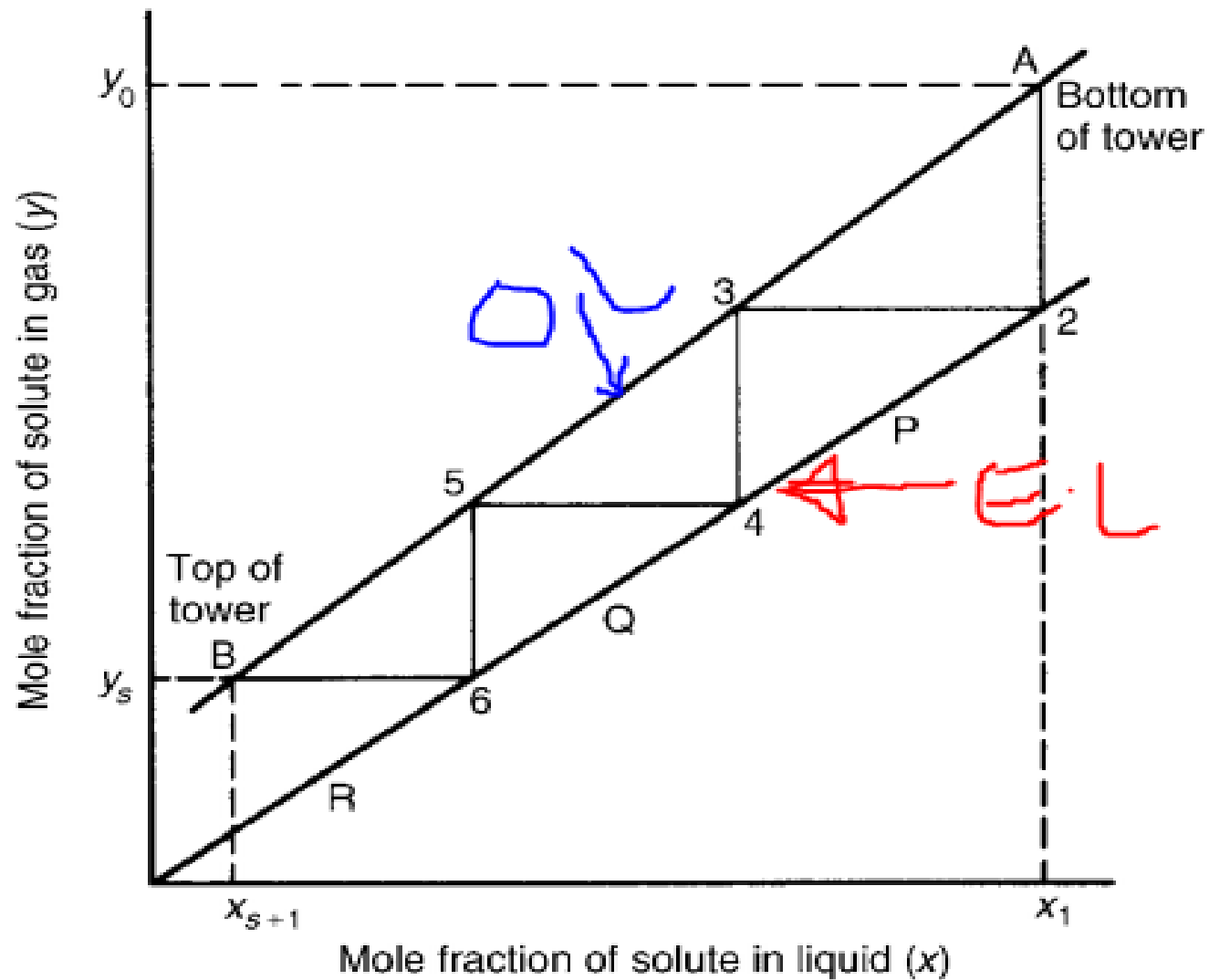


Figure 12.27. Diagrammatic representation of changes in a plate column

- As shown in Figure 12.27, such a line passes through two points $B(x_{s+1}, y_s)$ and $A(x_1, y_o)$, representing the terminal concentrations in the column. The equilibrium curve is shown in this figure as **PQR**.
- Point A represents conditions at the bottom of the tower.
- The gas rising from the bottom plate is in equilibrium with a liquid of **concentration x_1** and is shown as point 3 on the operating line.
- Then point **4** indicates the concentration of the liquid on the second plate from the bottom.
- In this way steps may be drawn to point **B**, giving the gas y_s rising from the top plate and the liquid x_{s+1} entering the top of the absorber.

Advantages of Trayed Columns

- 1) Plate columns can handle a wider range of liquid and gas flow-rates than packed columns.
- 2) Packed columns are not suitable for very low liquid rates.
- 3) The efficiency of a plate can be predicted with more certainty than the equivalent term for packing (HETP or HTU).
- 4) Plate columns can be designed with more assurance than packed columns. There is always some doubt that good liquid distribution can be maintained throughout a packed column under all operating conditions, particularly in large columns.
- 5) It is easier to make cooling in a plate column; coils can be installed on the plates.
- 6) It is easier to have withdrawal of side-streams from plate columns.
- 7) If the liquid causes fouling, or contains solids, it is easier to provide cleaning in a plate column; manways can be installed on the plates. With small diameter columns it may be cheaper to use packing and replace the packing when it becomes fouled.

Minimum liquid to gas ratio for absorbers

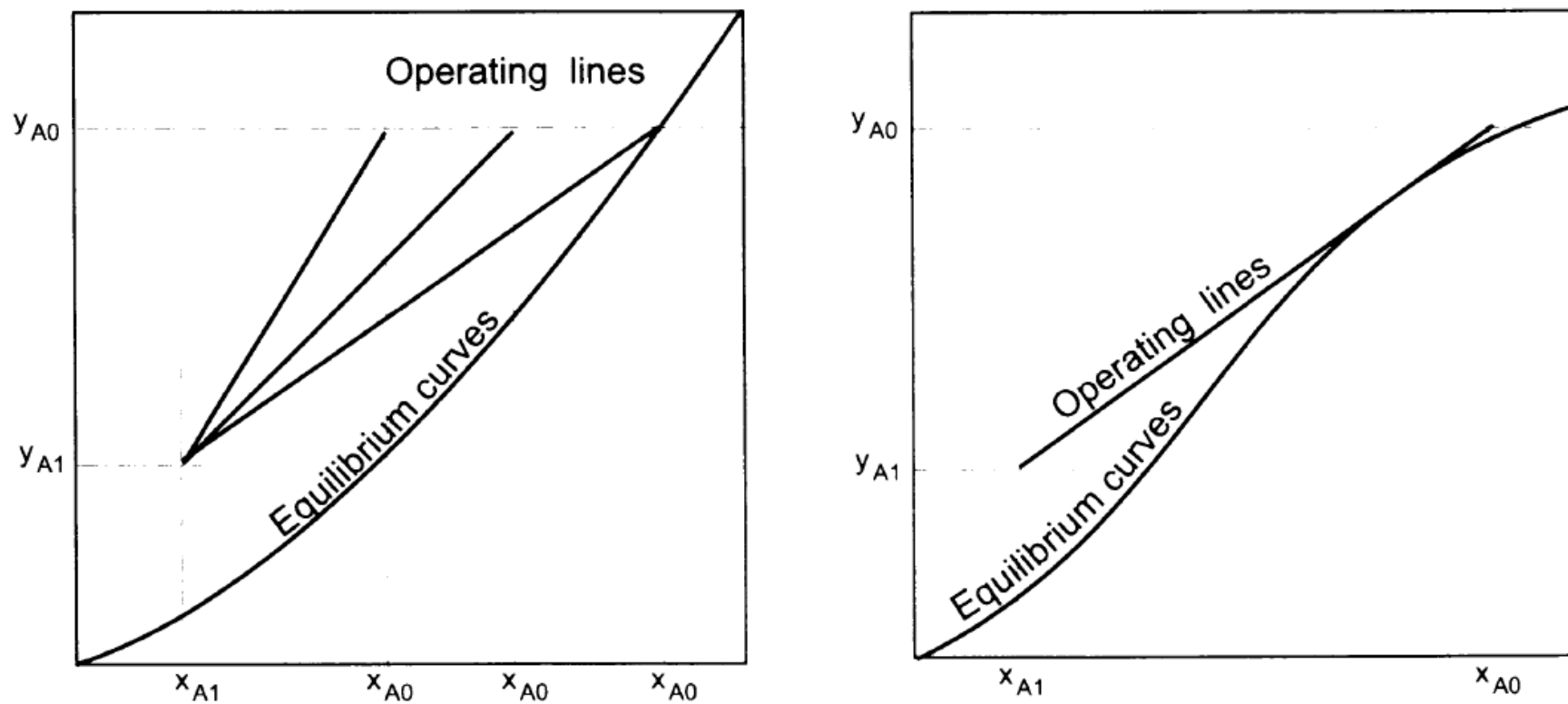


FIGURE 11.11 Graphical absorption tower relations.

- With a concave equilibrium line (Fig 11.11 left panel), three possible operating lines are shown. L/G is the slope of the operating line.
- The line furthest to the right with the smallest slope represents the limit to which the slope of the operating line can be drawn.
- Thus, this slope represents a minimum L/G ratio in which x_{A0} and y_{A0} are actually in equilibrium, and the line could not be drawn with a smaller slope because it would go through equilibrium.

- Since true thermodynamic equilibrium can never be reached practically, this point represents a theoretical minimum L/G which never can be attained in practice.
- Real L/G ratios are usually set at 1.1 to 1.7 times this minimum ratio.
- With “S”-shaped equilibrium curve (Fig. 11.11), $(L/G)_{\min}$ is reached at some point within the column where the operating and equilibrium lines are tangential and thermodynamic equilibrium would be “exceeded” if the L/G ratio were any smaller.

End